MINI PROJECT REPORT

# Distributed Consensus Simulator

## Raft-Style Distributed Systems Simulation Dashboard

## 📚 Index


| Sr. No. | Topic Name | Page |
|---|---|---|
| 1. | Introduction & Objectives | 1 |
| 2. | System Architecture & Raft Model | 2 |
| 3. | Dashboard & User Interface Screenshot | 3 |
| 4. | Control Panel Modules & Operations | 4 |

# 1. Introduction & Objectives

**Distributed Consensus Simulator** is an advanced web-based project designed to understand data consistency and node coordination processes in Distributed Systems. This project is based on the **Raft Consensus Algorithm**, which is widely used in modern cloud architectures and blockchain backends.

**Key Objectives of the Project:**

• Simulate real-time data proposals and voting processes among multiple independent nodes.
• Test system fault tolerance during node failure and recovery scenarios.
• Provide users with an interactive control panel to dynamically add, remove, and elect leaders across network nodes.

# 2. System Architecture

This project operates on a client-side interactive architecture utilizing the State Machine Replication model. The system is primarily composed of three core components:

**1. Node Cluster Manager:** Monitors the real-time status (Online, Offline, Leader, Follower) of all active nodes.
**2. Consensus Engine:** Computes majority voting mechanics for data proposals.
**3. Live Event Logger:** Displays every system event and timestamp on the screen in real-time.

# 3. Dashboard Interface

The image below provides a live view of the main dashboard of our **Distributed Consensus Simulator**, showcasing the control panel, node status cards, and live event log.

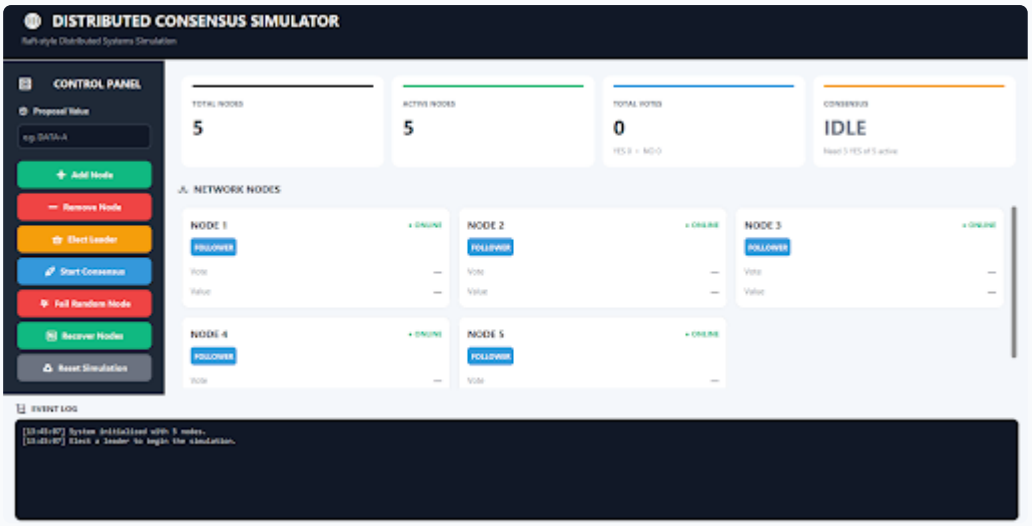



*Figure 3.1: Complete view of the Distributed Consensus Simulator Dashboard and Control Panel*

**Dashboard Features:** Metric cards at the top display total nodes, active nodes, total votes, and consensus status. In the network nodes section, the role (Follower/Leader) and status of each individual node are clearly visible.

## 4. Control Panel Modules & Operations

The control panel on the left side of the project provides several powerful options to manage and interact with the simulation system:

### Core Functions:

• **Add Node:** Adds a new node to the active network.
• **Remove Node:** Removes a selected node from the network.
• **Elect Leader:** Initiates leader election among nodes using the Raft algorithm.
• **Start Consensus:** Triggers the voting process for a data proposal.

### Fault & Reset Options:

• **Fail Random Node:** Simulates a node failure to test fault tolerance.
• **Recover Nodes:** Brings failed nodes back online.
• **Reset Simulation:** Resets the entire simulation to its initial state.

## Conclusion

Through this project, the practical implementation of how consensus is achieved in distributed systems was successfully demonstrated. The clean UI, real-time event logs, and robust control panel make this project highly engaging and educational.